Ari Ofogh

Washington, D.C. 20037 shapari.ofogh@gwu.edu (804)-510-6194 linkedin.com/shapari-ofogh/

EDUCATION

The George Washington University, School of Engineering and Applied Science Washington, D.C.
Bachelor of Science in Mechanical Engineering Expected Graduation: December 2027

- **Certifications:** Certified SolidWorks CAD Associate (CSWA), LinkedIn Microsoft Project Step by Step: Planning for Successful Project Management, LinkedIn Reverse Engineering Foundations: Product Design
- **Relevant Coursework:** Thermal Systems Design, Fluid Mechanics, Heat Transfer, Mechanical Systems Design, Manufacturing Process & Systems, Engineering Drawing & Computer Graphics

SKILLS

- **Mechanical Modeling:** SolidWorks | AutoCAD | Fusion 360
- **Finite Element Analysis (FEA):** SolidWorks Simulation
- **Simulation:** MATLAB Simulink
- **Languages:** English, Persian, & Conversational Turkish
- **Manufacturing:** 3D Printing | Lathe & Mill | GD&T
- **Programming Languages:** Python | MATLAB
- **Productivity:** Microsoft Office | Notion

EXPERIENCE

Assistive Robotics and Telemedicine (ART-Med) Lab Washington, D.C.
Undergraduate Research Assistant October 2024 - Present

- Focusing research on integrating human intelligence with robotics to support adolescents with disabilities, both socially and emotionally
- Implement tactile sensor mapping and haptic feedback integration using Python, tactile sensors, and robotic glove control software to enable teleoperated robotic painting and manipulation.
- Designed and fabricated an Ionic Polymer-Metal Composite (IPMC) actuator, integrating material processing electrode deposition, and electromechanical testing to evaluate displacement response and performance characteristics
- Developed a robotic head prototype for the Go2 Robot Dog to promote socio-emotional interactions and individualized assistance for children with autism spectrum disorder
- Presenting research findings regularly at weekly lab meetings to update PI and peer research assistants

George Washington University Biomedical Engineering Society (BMES) Washington, D.C.
Vice President of Operations & Finance January 2025 – Present

- Managing the club's budget by tracking expenses and updating spreadsheets
- Requesting funding from student government through formal proposals to secure money for events and resources the club needs; handling reimbursements and payments
- Giving weekly financial updates to the executive board to keep everyone informed and support quick decision-making

Zaytinya, José Andrés Group Washington, D.C.
Hostess August 2026 – Present

- Delivered a welcoming first impression and ensured positive guest experiences from arrival to departure; managed reservations, waitlists, and table assignments to optimize seating flow and guest comfort
- Communicated with servers, bussers, and kitchen staff during peak service periods; organized and packaged to-go orders

PROJECTS

Functionally Graded Materials for Thermal Management
Senior Capstone Design | Lockheed Martin & NSWC

- Designing functionally graded material (FGM) stacks to improve thermal management and minimize thermal stresses in aerospace and defense applications
- Analyzing metal-ceramic and metal-metal combinations based on thermal conductivity, thermal expansion, and Young's modulus using finite element analysis (FEA) models such as COMSOL and ABAQUS

Electromechanical System Design Project
Team Lead

- Led a 3-member team through a 2-month engineering challenge to design and build a timed mechanical subsystem controlling a golf ball for exactly 15 seconds within a 12-team system-of-systems loop within a $225 budget
- Personally designed, hands-on prototyped, and troubleshooted a camshaft-driven lifting mechanism using bevel gears and oscillating inclined columns, leading the full engineering workflow from concept sketches and SolidWorks CAD modeling to fabrication, torque analysis, and FEA validation
- Programmed an Arduino Uno to control motor movement and timing using a motor driver circuit